WEEK 4 – ASSIGNMENT

Amrita Ruhela CT_CSI_DE_1102 DIT University

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

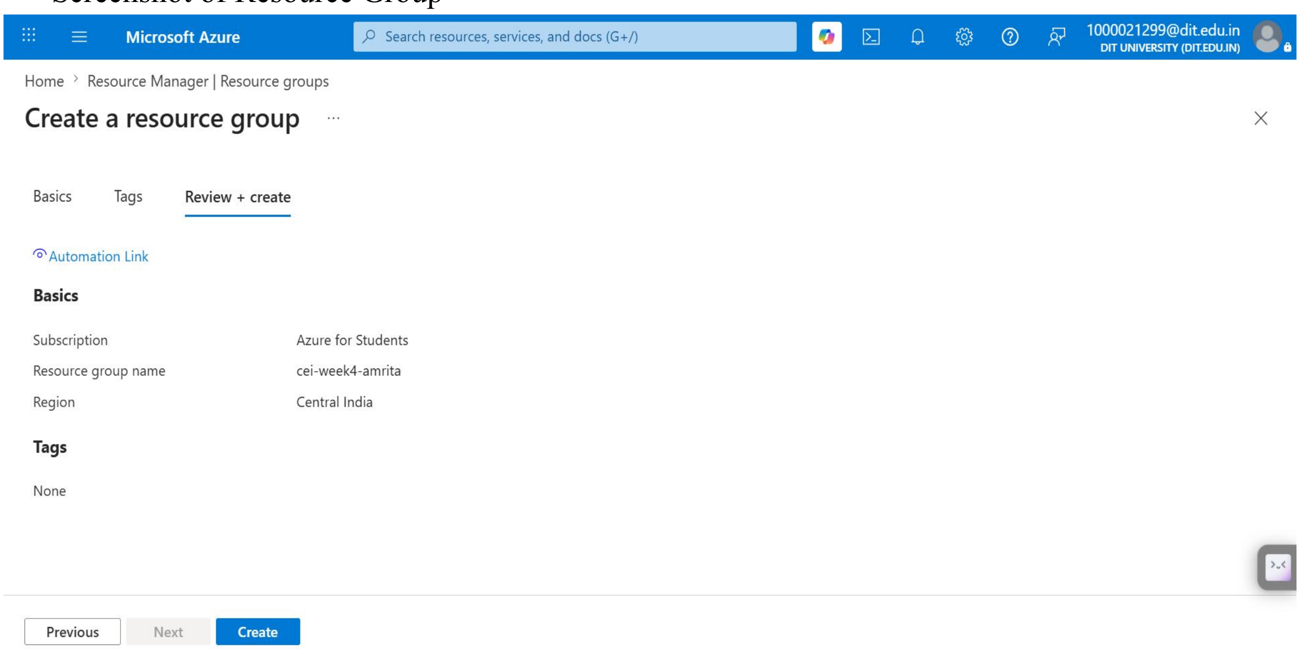
Assignment Tasks

Task 1:

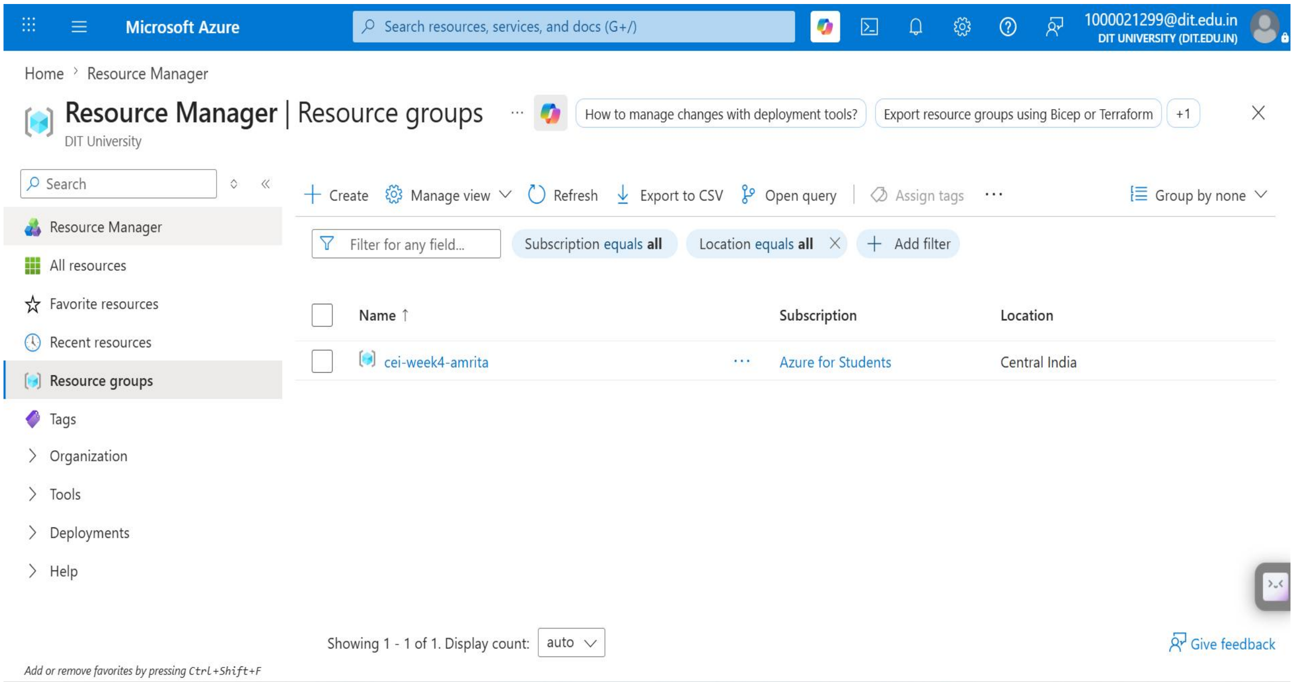
- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



WEEK 4 - ASSIGNMENT



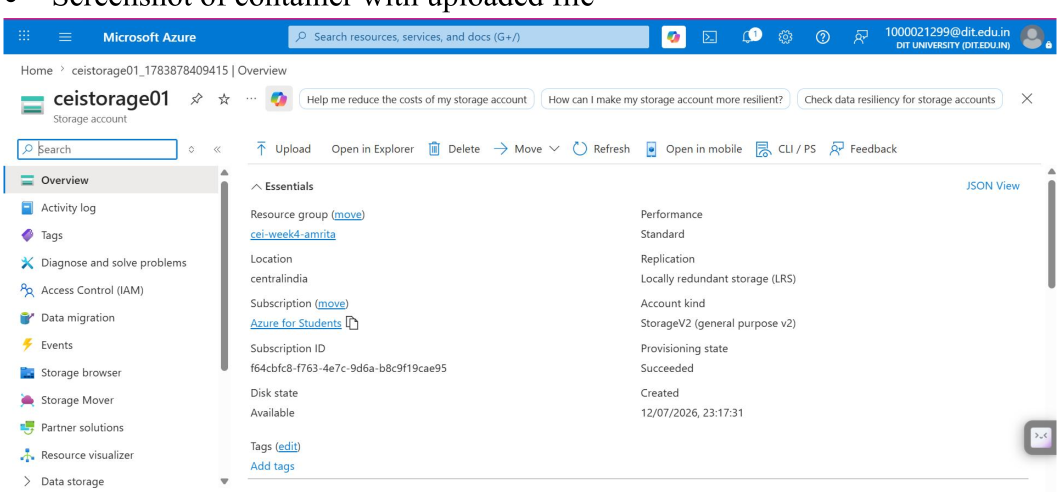
Task 2: Storage Setup

Do:

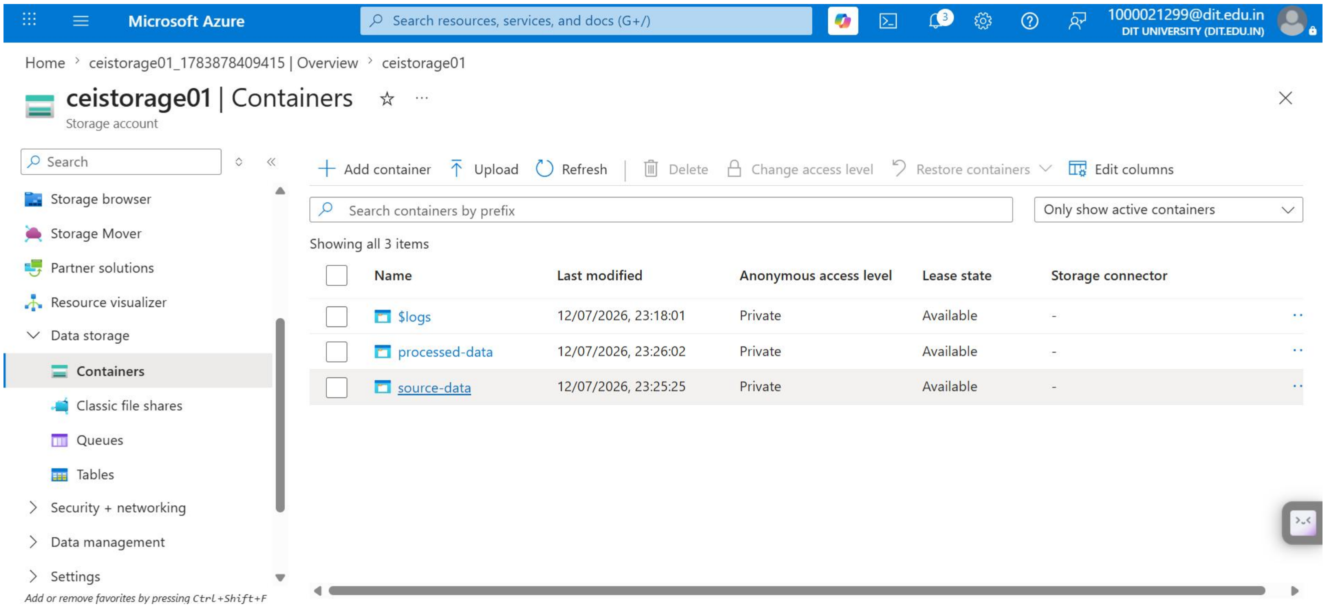
- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

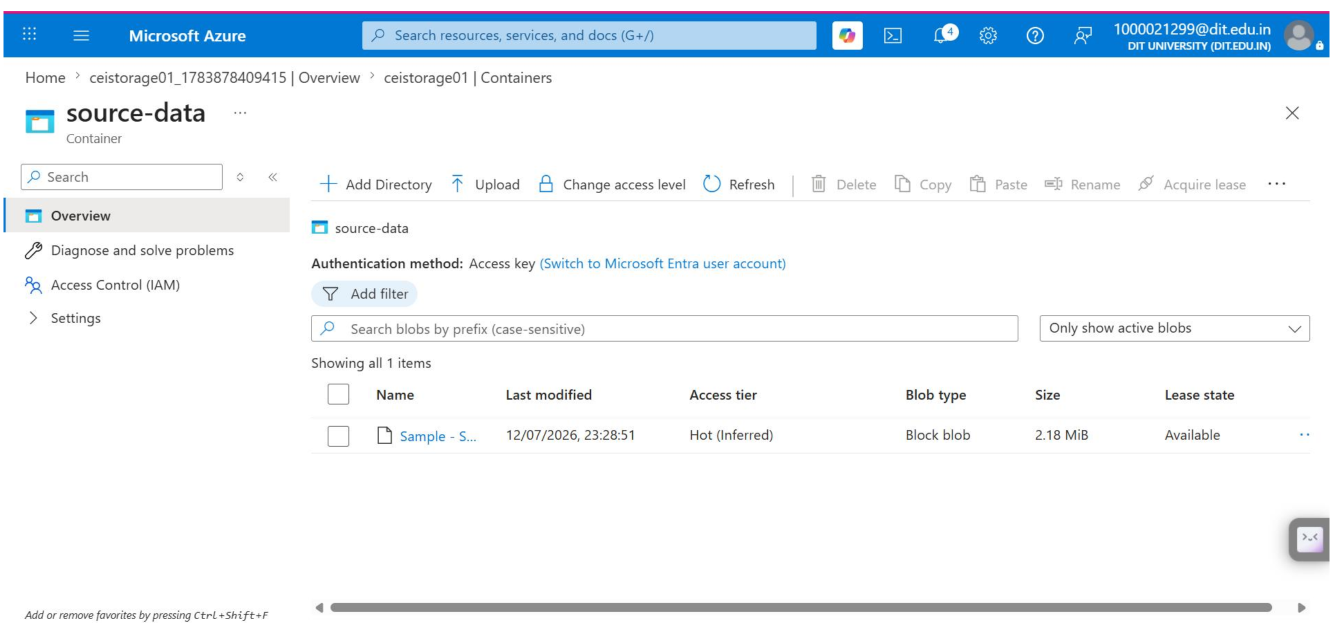
Deliverable:

- Screenshot of container with uploaded file



WEEK 4 - ASSIGNMENT





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Task 3: ADF Basics

Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

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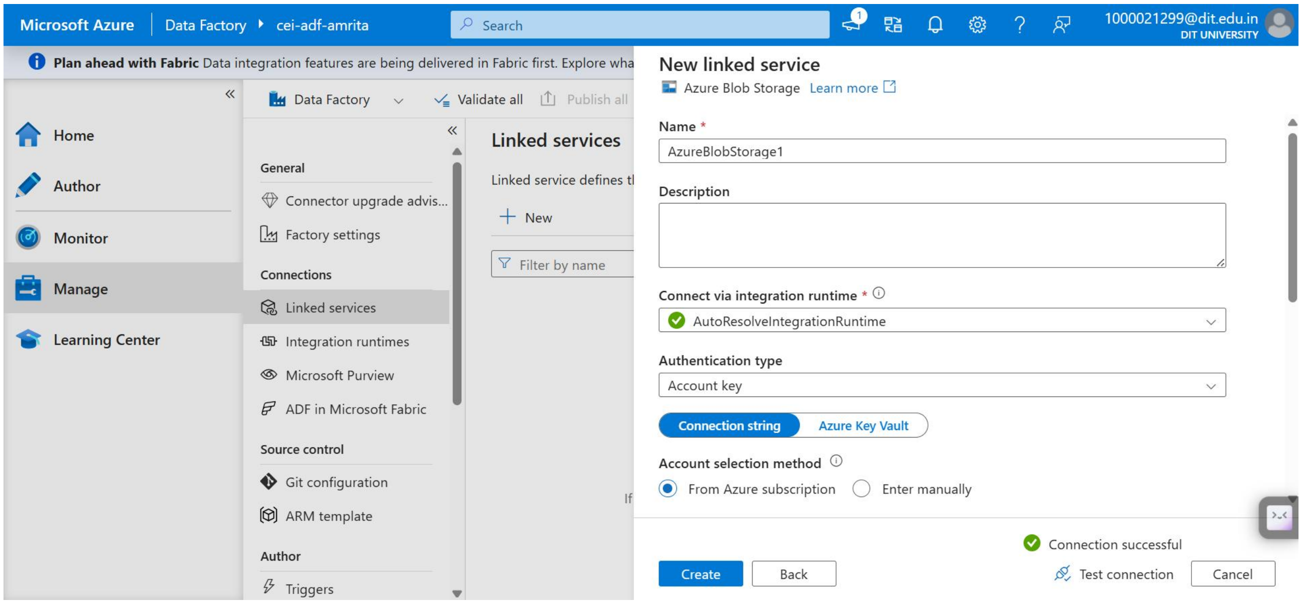
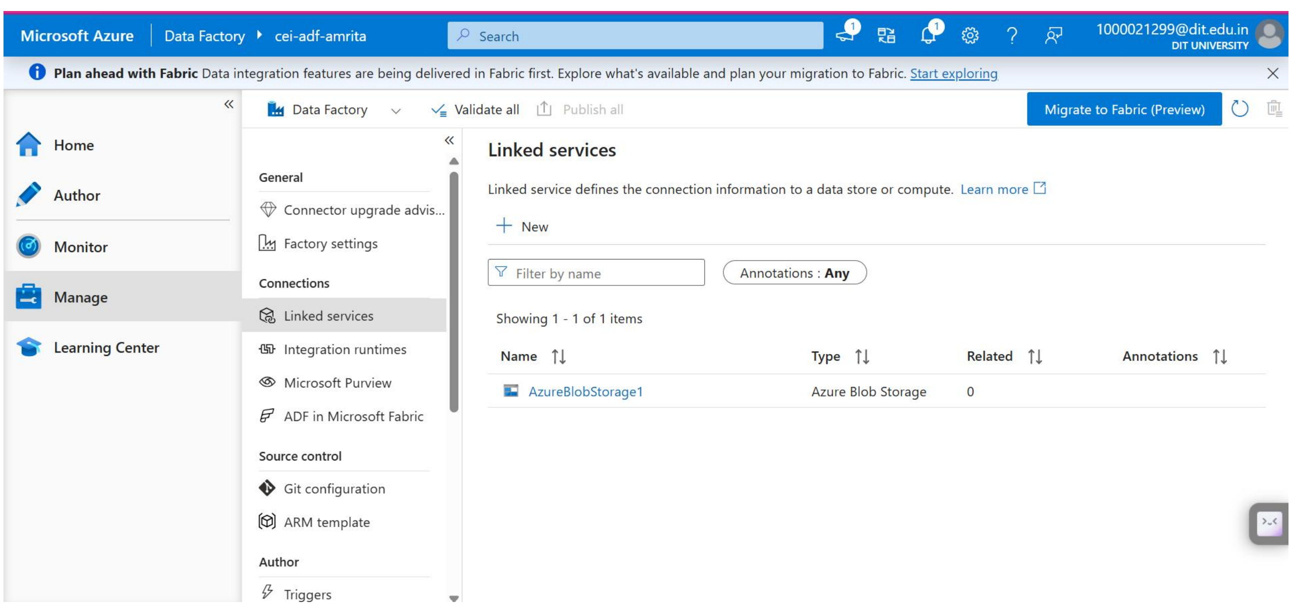
- Screenshot of Linked Service





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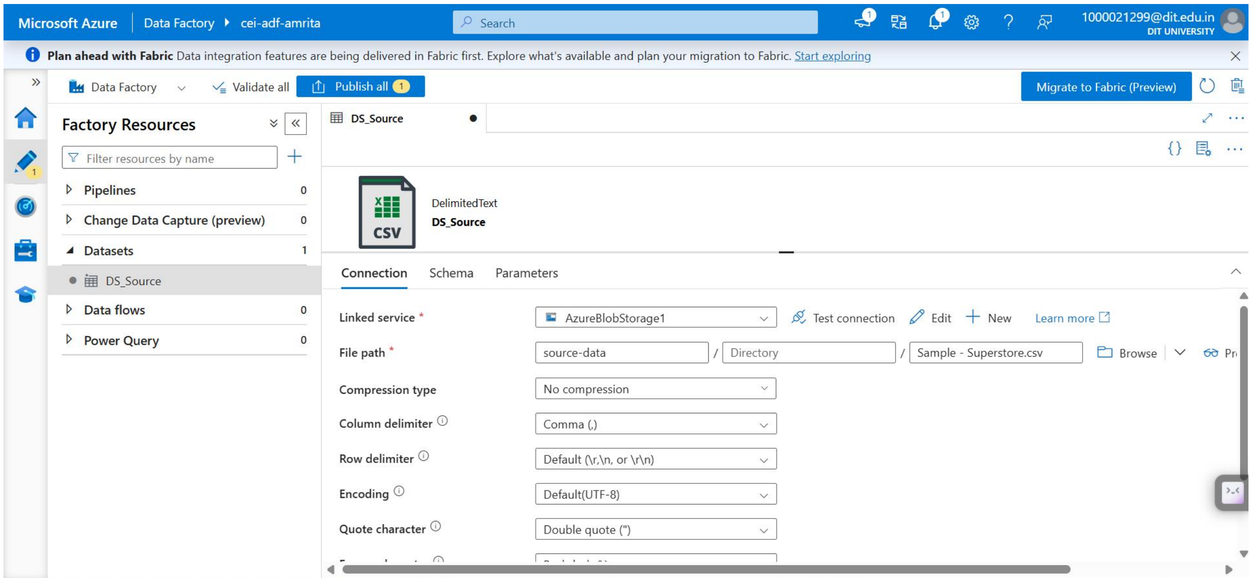
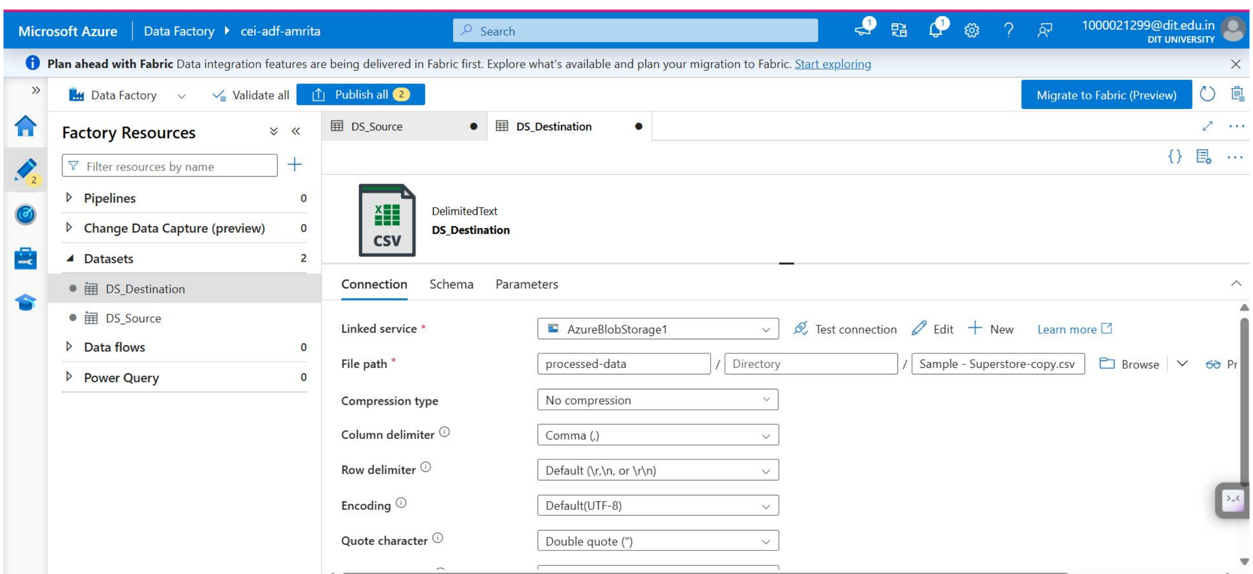
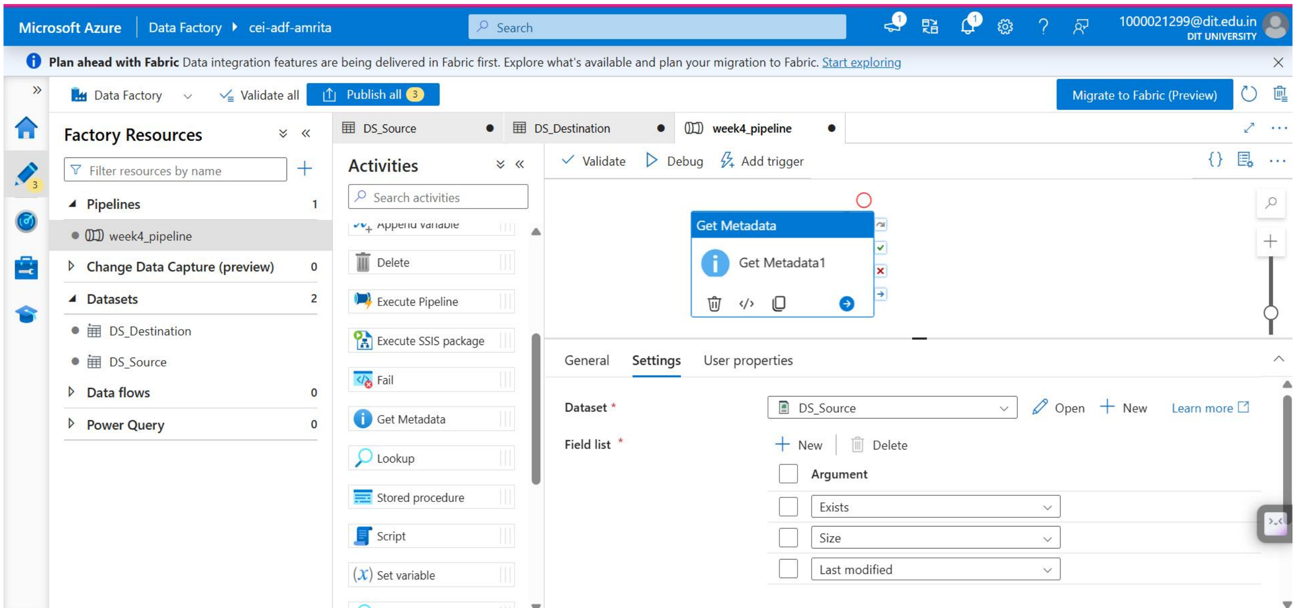
- Screenshot of dataset





WEEK 4 - ASSIGNMENT

- Screenshot of Get Metadata activity



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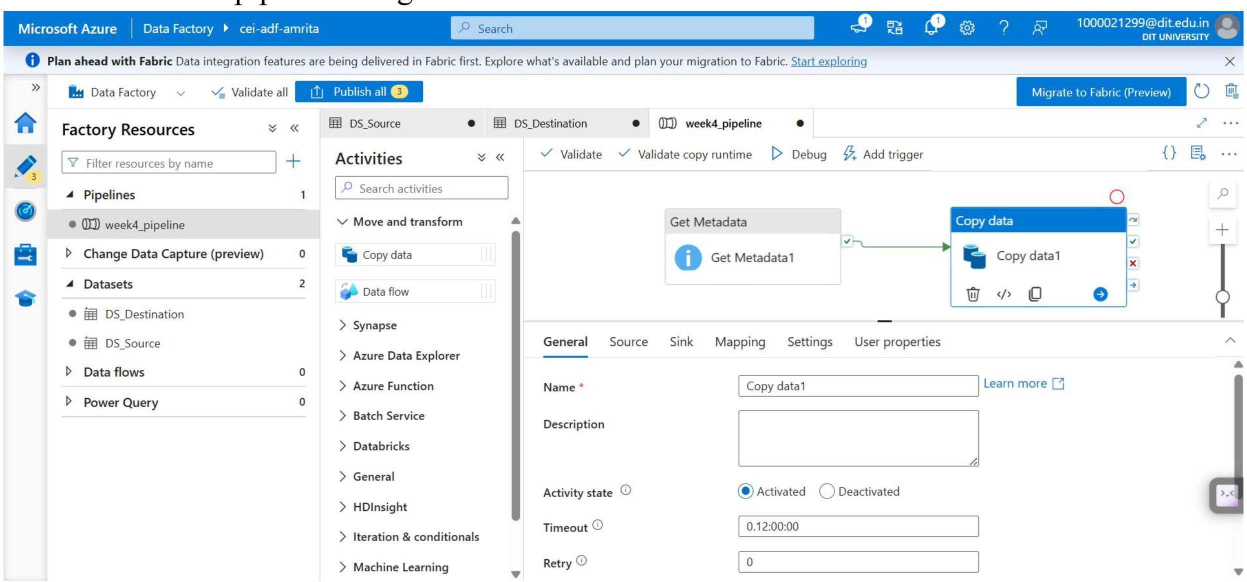
Task 4: Pipeline Development

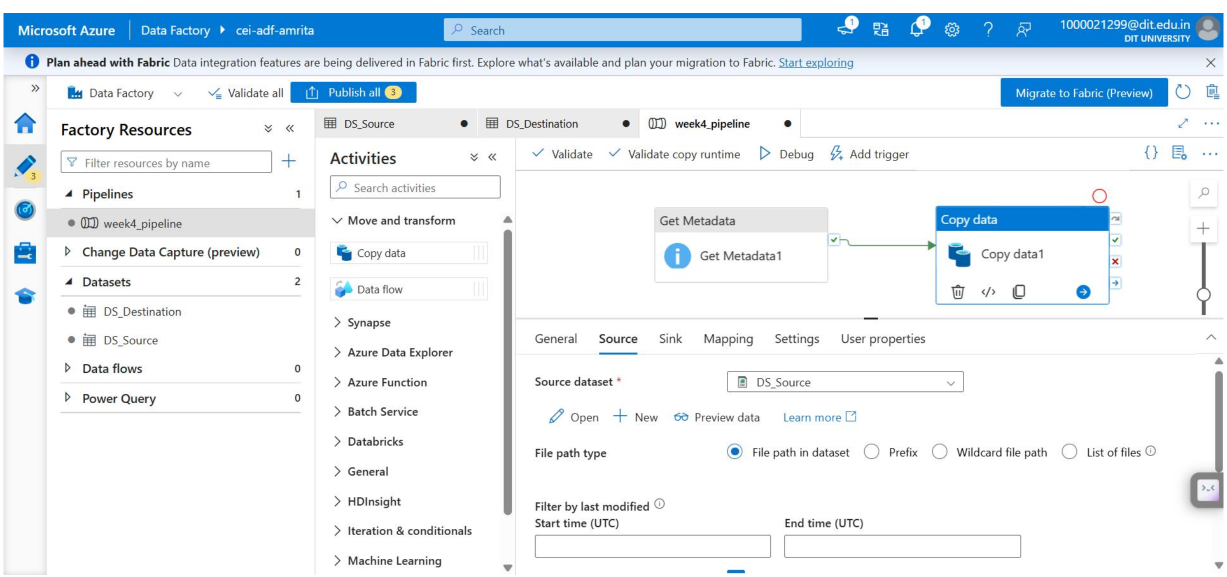
Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

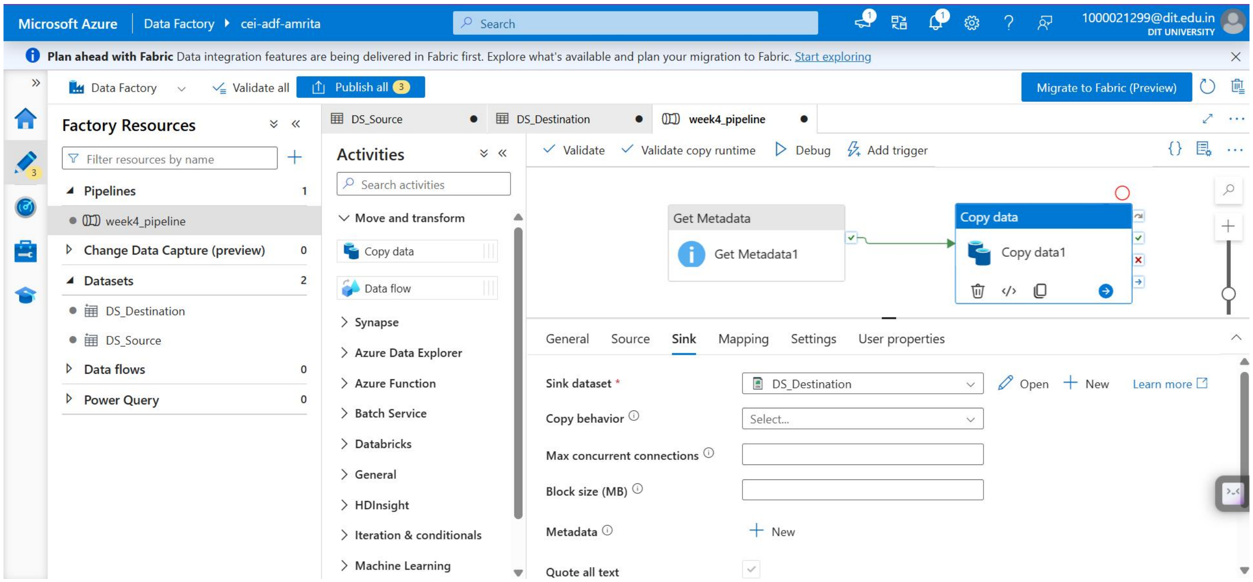
Deliverable:

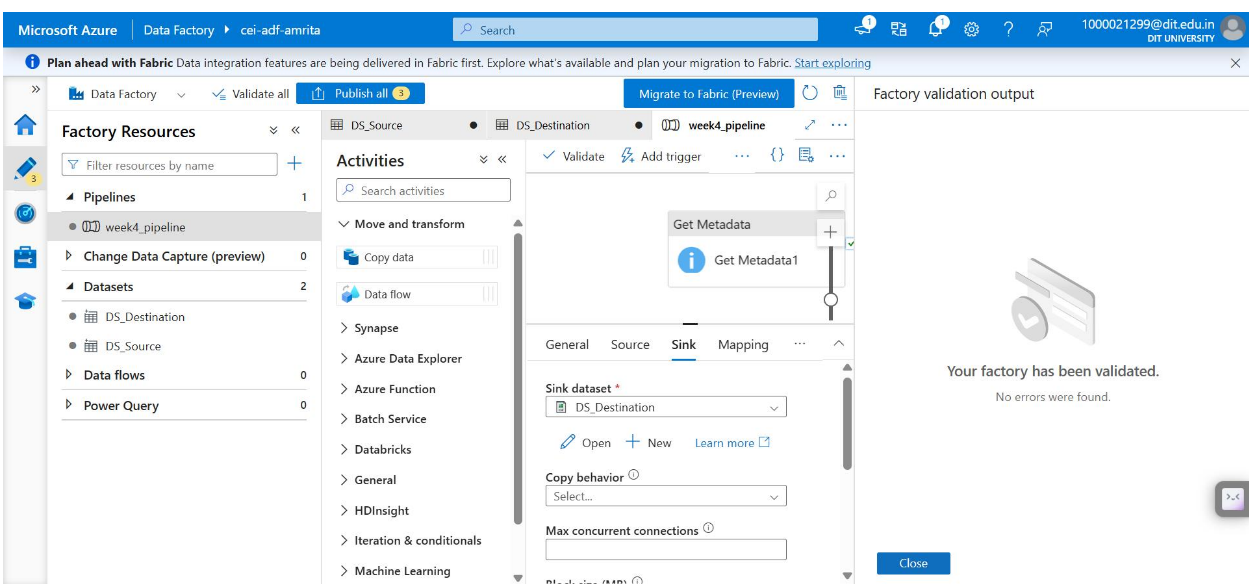
- Screenshot of pipeline design





WEEK 4 - ASSIGNMENT





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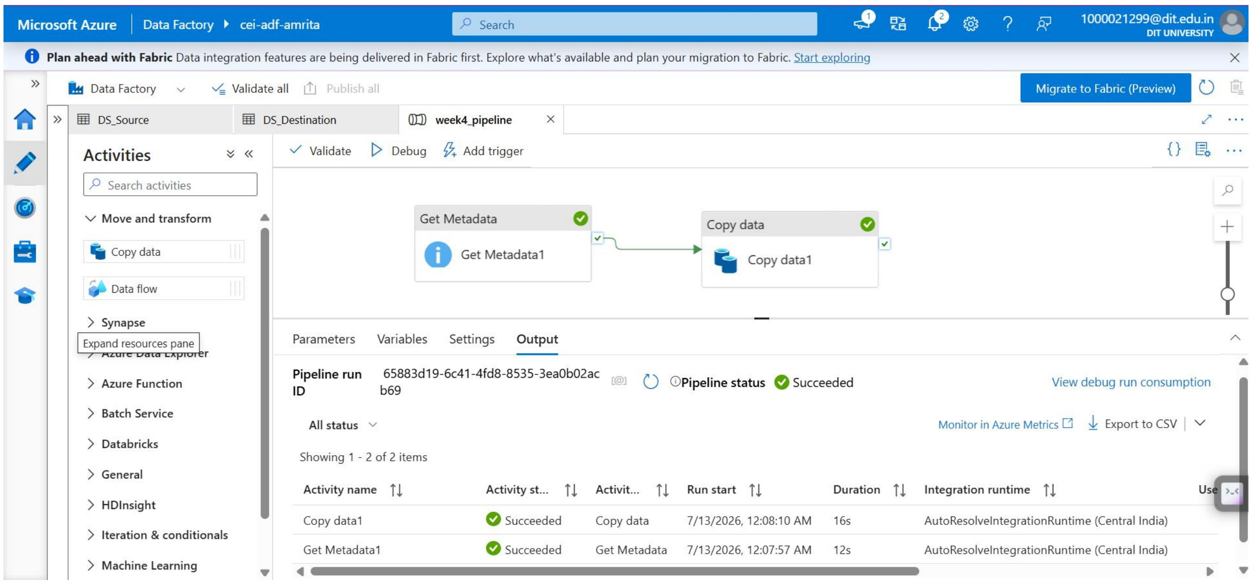
Task 5: Pipeline Execution

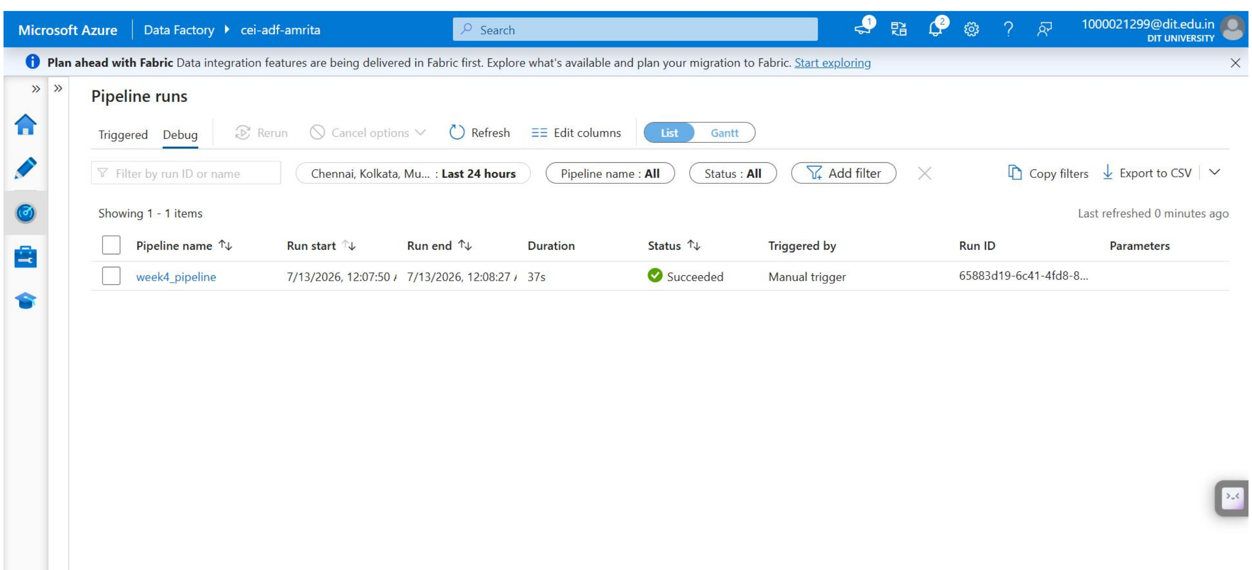
Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)





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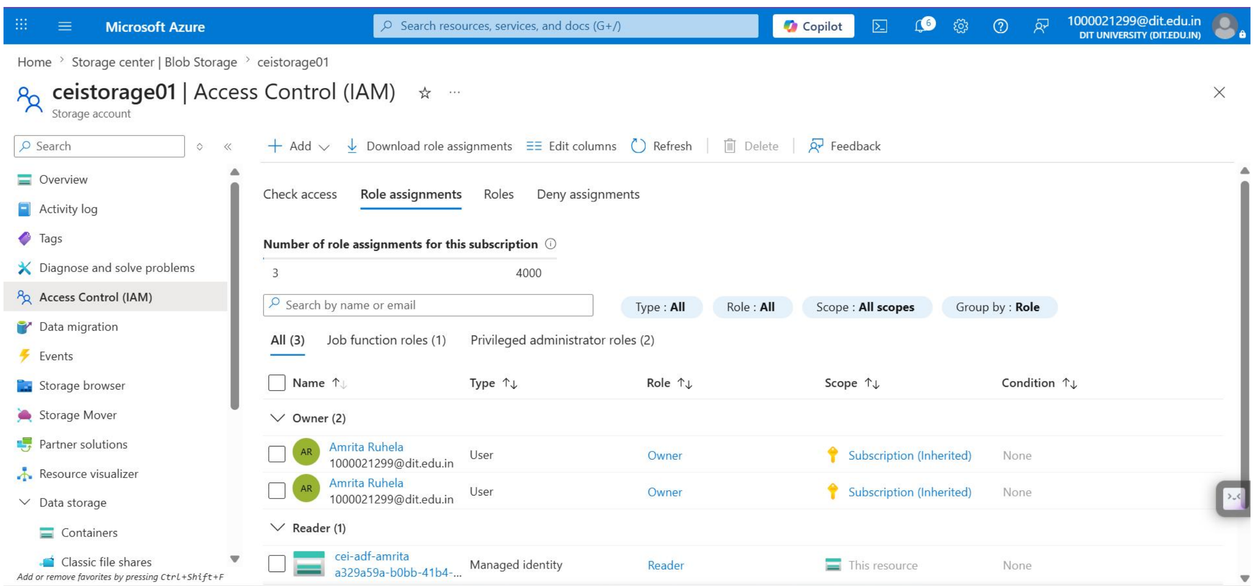
Task 6: IAM Roles

Do:

- Assign roles:
  - Reader
  - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment



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Mini Project (Final Task)

Problem Statement:

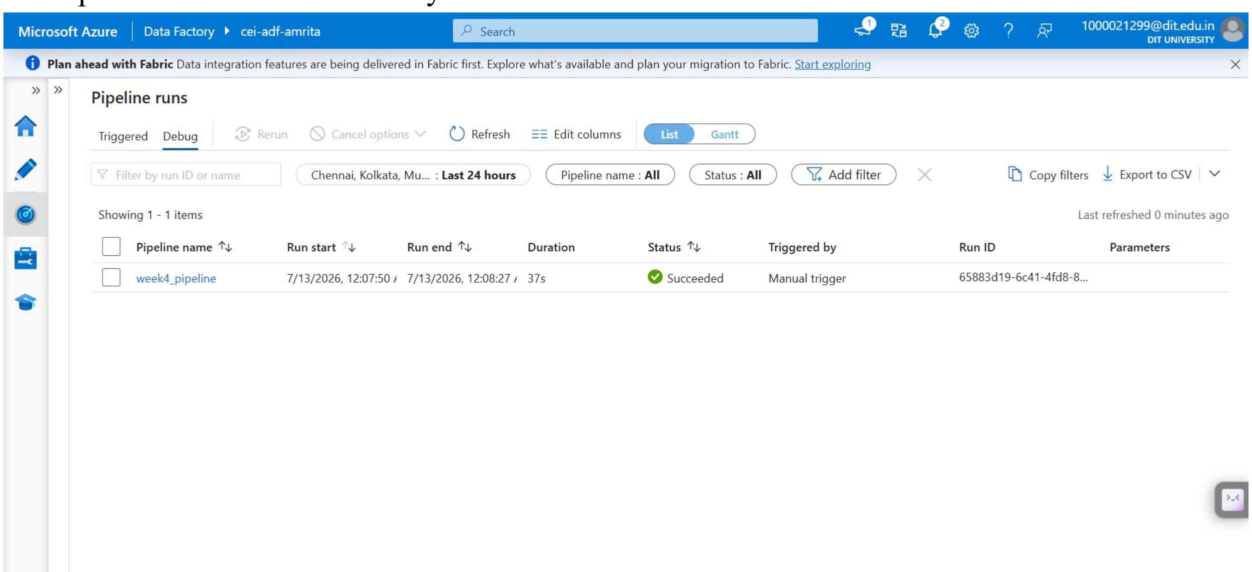
Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully



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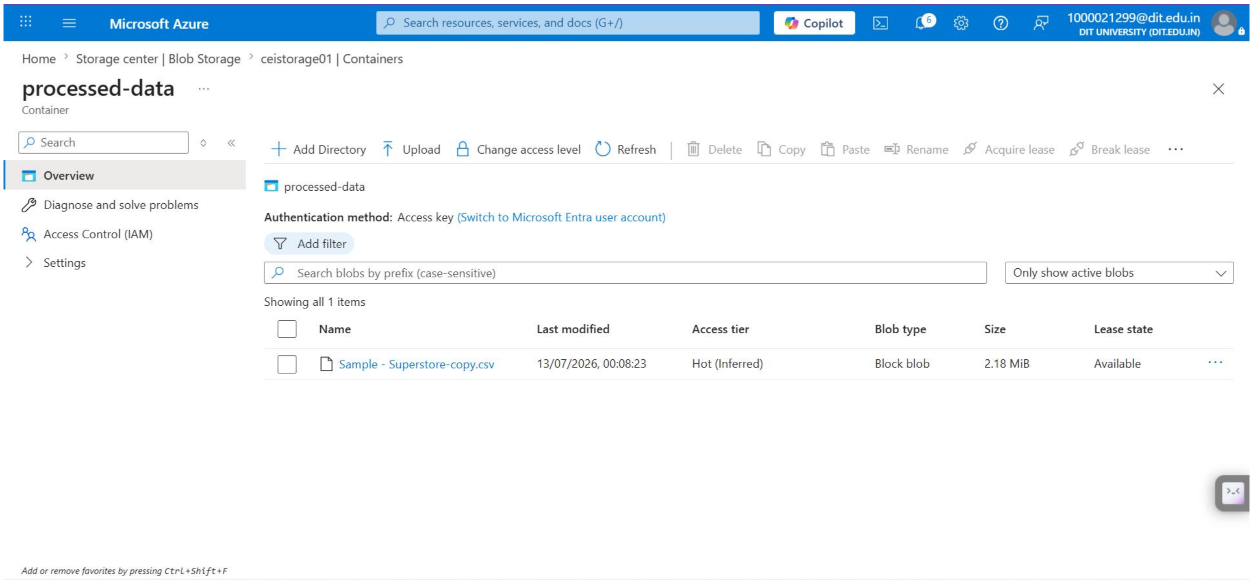
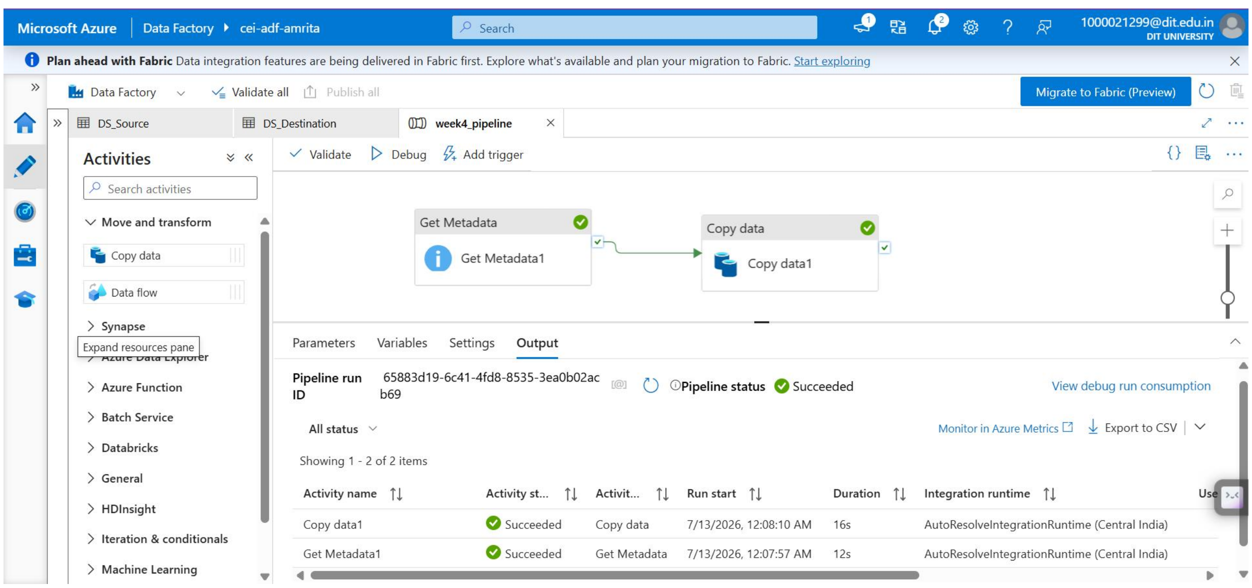
- Data copied to destination



- Metadata validated



WEEK 4 - ASSIGNMENT

Summary:
In this project, an Azure Resource Group, Storage Account, Blob Storage containers, and Azure Data Factory were successfully created and configured. Linked Services and Datasets were established to connect Azure Data Factory with Azure Blob Storage. A pipeline was developed using the Get Metadata activity to validate the source file's existence, size, and last modified timestamp, followed by the Copy Data activity to transfer the CSV file from the source container to the destination container. Appropriate IAM roles (Reader and Contributor) were assigned to ensure secure access between Azure Data Factory and Azure Storage. The pipeline was validated, published, executed successfully, and the output was verified by confirming the copied file in the destination container along with successful metadata validation.

Outcome:
- Successfully created and configured Azure cloud resources.
- Connected Azure Blob Storage with Azure Data Factory using Linked Services.
- Validated source file metadata before data movement.
- Successfully copied the CSV file to the destination container.
- Verified pipeline execution and destination output.
- Demonstrated an end-to-end Azure Data Factory ETL workflow using Azure cloud services.